

Lillian Barrett Hughes Wyatt

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California Institute of Technology, 1200 East California Blvd. MC 104-44, Pasadena, CA 91125 U.S.A.

RESEARCH VISION

Engineering materials that enable robust quantum coherence and controlled interactions by integrating synthesis, characterization, and fabrication, with quantum measurements.

PROFESSIONAL EXPERIENCE

• California Institute of Technology

AWS Quantum Postdoctoral Scholar

Nov. 2025 - Present

Pasadena, CA, U.S.A.

EDUCATION

• University of California Santa Barbara

Ph.D. in Materials

Advisors: Professor Ania C. Bleszynski Jayich and Professor Kunal Mukherjee (Stanford)

Sep. 2019 - Sep. 2025

Santa Barbara, CA, U.S.A.

• University of Richmond

B.S. in Interdisciplinary Physics, concentration in Chemistry

Summa cum laude

Advisor: Professor Michael C. Leopold

Aug. 2015 - May 2019

Richmond, VA, U.S.A.

HONORS AND AWARDS

- **Materials Research Society Graduate Student Silver Award**, MRS Spring Meeting (2025)
- **Best Student Oral Presentation Award**, MRS Fall Meeting, Diamond Symposium (2023)
- **First Place Student Presentation Award**, New Diamond and Nano Carbon Conference (2023)
- **Best Student Oral Presentation Award**, European MRS Meeting, Diamond Symposium (2023)
- **UCSB Materials Department Service Award**, Bright Horizon Global Foundation (2022)
- **NSF Graduate Research Fellowship** (2021)
- **NDSEG Fellowship** (2021)
- **UCSB NSF Quantum Foundry Graduate Fellowship** (2021)
- **NSF GRFP Honorable Mention** (2019)
- **UC Regents Fellowship**, Materials Department, UCSB (2019)
- **David C. Evans Award for Outstanding Achievement in Scholarship**, College of Arts and Sciences, UR (2019)
- **Robert E. Loving Award for Top Graduating Student**, Department of Physics, UR (2019)
- **Goldwater Scholarship** (2018)
- **Phi Beta Kappa**, UR (2018)
- **Multiple departmental awards in Physics and Chemistry**, UR (2017–2018)

PUBLICATIONS

- [1] E. Postelnicu, **L. B. Hughes Wyatt**, T. Ngyuen, S. A. Meynell, C. Jilly, P. Wallace, A. Barnum, A. C. Bleszynski Jayich, K. Mukherjee. Harnessing diamond surface features for dense and aligned NV ensembles. *Nano Lett.* (2026). doi.org/10.1021/acs.nanolett.6c02155
- [2] C. C Newsom, **L. B Hughes Wyatt**, B. L Green, A. C Bleszynski Jayich, M. E Newton. 200 keV energy electron irradiation of single crystal diamond: Quantification of vacancy and nitrogen-vacancy production. *Phys. Rev. B.* **114**, 065203 (2026). doi.org/10.1103/hls1-92jc
- [3] H. Gao, N. T. Leitaio, S. Dandavate, **L. B. Hughes Wyatt**, P. Put, M. Mammen, L. S. Martin, H. Park, A. C. Bleszynski Jayich, M. D. Lukin. Dressed-state Hamiltonian engineering in a strongly interacting solid-state spin ensemble. *Phys. Rev. Lett.* **136**, 200802 (2026). doi.org/10.1103/d1kx-hy93
- [4] **L. B. Hughes Wyatt**,* S. Parthasarathy,* I. Kantor,* C. K. Kim, L. Chen, T. A. Morrison, J. Ahlers, K. Mukherjee, A. C. Bleszynski Jayich. Creation of depth-confined, shallow nitrogen-vacancy centers in diamond with tunable density. *Appl. Phys. Lett.* **128**, 204006 (2026). doi.org/10.1063/5.0316283
- [5] H. Lee,* H. C. Kleidermacher,* A. J. M. Stein,* H. Oh, **L. B. Hughes Wyatt**, C. K. Kim, L. Basso, A. Mounce, S. S. Su, M. Titze, A. C. Bleszynski Jayich, J. Vuckovic. Quantum nanophotonic interface for tin-vacancy centers in thin-film diamond. *Phys. Rev. Appl.* **25**, 044074 (2026). doi.org/10.1103/vwt7-b64x
- [6] H. Oh,* V. Dharod,* C. Padgett,* **L. B. Hughes Wyatt**, J. Venkatraman, S. Parthasarathy, E. Osipova, I. Hedgpeeth, J. V. Cady, L. Basso, Y. Wang, M. Titze, E. S. Bielejec, A. M. Mounce, D. Bouwmeester, A. C. Bleszynski Jayich. Spin-embedded diamond optomechanical resonator with a mechanical quality factor exceeding one million. *Optica* **13**, 485-490 (2026). doi.org/10.1364/OPTICA.577541

- [7] B. Edelman, C. Sheppard, L. Chuidian, A. Vinnikov, F. Bevc, **L. B. Hughes**, K. Kittredge, C. Parish, M. Leopold. Design of Nanomaterial-Based Sensors for Enhanced Halogen Bonding. *ACS Omega* **11**, 4006-4020 (2026). doi.org/10.1021/acsomega.5c07542
- [8] S. Kim, P. London, D. Yang, **L. B. Hughes**, J. Ahlers, S. Meynell, W. J. Mitchell, K. Mukherjee, A. C. Bleszynski Jayich. Scalable nanoscale positioning of highly coherent color centers in prefabricated diamond nanostructures. *Nat. Comm.* **16**, 9803 (2025). doi.org/10.1038/s41467-025-64758-4
- [9] H. Gao, L. S. Martin, **L. B. Hughes**, N. T. Leitao, P. Put, B. Ye, H. Zhou, W. Wu, N. U. Koyluoglu1, S. Meynell, E. Davis, N. Yao, A. C. Bleszynski Jayich, H. Park, M. D. Lukin. Signal amplification in a solid-state quantum sensor via asymmetric time-reversal of many-body dynamics. *Nature* **646**, 68-74 (2025). doi.org/10.1038/s41586-025-09452-7
- [10] W. Wu, E. J. Davis, **L. B. Hughes**, B. Ye, Z. Wang, D. Kufel, T. Ono, S. A. Meynell, M. Block. C. Liu, H. Yang, A. C. Bleszynski Jayich, N. Y. Yao. Spin squeezing in an ensemble of nitrogen-vacancy centers in diamond. *Nature* **646**, 74-80 (2025). doi.org/10.1038/s41586-025-09524-8
- [11] **L. B. Hughes**, S. A. Meynell, W. Wu, S. Parthasarathy, L. Chen, Z. Zhang, Z. Wang, E. J. Davis, K. Mukherjee, N. Y. Yao, and A. C. Bleszynski Jayich. A Strongly Interacting, Two-Dimensional, Dipolar Spin Ensemble in (111)-Oriented Diamond. *Phys. Rev. X* **15**, 021035 (2025). doi.org/10.1103/PhysRevX.15.021035
- [12] S. Parthasarathy, M. Joos, **L. B. Hughes**, S. A. Meynell, T. Morrison, K. Mukherjee, D. Weld, A. C. Bleszynski Jayich. Role of oxygen in laser induced contamination of diamond-vacuum interfaces. *Phys. Rev. Appl.* **22**, 024067 (2024). doi.org/10.1103/PhysRevApplied.22.024067
- [13] P. D. Reddy, L. Nordin, **L. B. Hughes**, A. K. Friedl, K. Mukherjee. Expanded stability of layered SnSe-PbSe alloys and evidence of displacive phase transformation from rocksalt in heteroepitaxial thin films. *ACS Nano* **18**, 13437-13449 (2024). doi.org/10.1021/acsnano.4c04128
- [14] L. V. H. Rogers, S. T. Nguyen, J. H. Cox, K. Zervas, Z. Yuan, S. Sangtawesin, A. Stacey, C. Jaye, C. Weiland, A. Pershin, A. Gali, L. Thomsen, S. A. Meynell, **L. B. Hughes**, A. C. Bleszynski Jayich, X. Gui, R. J. Cava, R. R. Knowles, N. P. de Leon. Diamond Surface Functionalization via visible light-driven CH activation for nanoscale quantum sensing. *PNAS* **121**, e2316032121 (2024). doi.org/10.1073/pnas.2316032121
- [15] **L. B. Hughes**, Z. Zhang, C. Jin, S. A. Meynell, B. Ye, W. Wu, Z. Wang, E. J. Davis, T. E. Mates, N. Y. Yao, K. Mukherjee, and A. C. Bleszynski Jayich. Two-dimensional spin systems in PECVD-grown diamond with tunable density and long coherence for enhanced quantum sensing and simulation. *APL Materials* **11**, 021101 (2023). doi.org/10.1063/5.0133501
*Selected as a featured article.
- [16] L. V. H. Rogers, **L. B. Hughes**, M. Xie, S. W. Kolkowitz, P. C. Maurer, A. C. Bleszynski Jayich, N. P. de Leon. Materials challenges for quantum technologies based on color centers in diamond. *MRS Bulletin* **46**, 623-633 (2021). doi.org/10.1557/s43577-021-00137-w
*Featured in a special collection of journal's most impactful publications.
- [17] S. A. Meynell, C. A. McLellan, **L. B. Hughes**, W. Wang, K. Mukherjee, and A. C. Bleszynski Jayich. Engineering quantum-coherent defects: the role of substrate miscut in chemical vapor deposition diamond growth. *Appl. Phys. Lett.* **117**, 194001 (2020). doi.org/10.1063/5.0029715
*Editor's pick.
- [18] **L. B. Hughes**,* N. Labban,* G. Conway, J. Pollock, and M.C. Leopold. Adaptable Xerogel-Layered Amperometric Biosensor Platforms on Wire Electrodes for Clinically Relevant Measurements. *Sensors* **19**, 2584 (2019). doi.org/10.3390/s19112584
- [19] A. K. Jaini, **L. B. Hughes**, M. Kitimet, K. J. Ulep, M. C. Leopold, and C. A. Parish. Halogen Bonding Interactions for Aromatic and Non-Aromatic Explosive Detection. *ACS Sensors* **4**, 389-397 (2019). doi.org/10.1021/acssensors.8b01246
- [20] P. Put,* N. T. Leitao,* H. Gao,* C. Spaegele,* O. Makarova, **L. B. Hughes**, A. C. Maccabe, M. Mammen, B. Machielse, H. Zhou, S. Pustelny, A. C. Bleszynski Jayich, F. Capasso, L. S. Martin, H. Park, M. D. Lukin. Collective many-body dynamics in a solid-state quantum sensor controlled through nanoscale magnetic gradients. *Accepted to PRX.* (2025) [arXiv:2506.11920](https://arxiv.org/abs/2506.11920)
- [21] Z. Zhang, T. Morrison, **L. B. Hughes**, W. Wu, R. Liu, D. Bluvstein, N. Yao, D. Fygenson, A. C. Bleszynski Jayich. Patterning programmable spin arrays on DNA origami for quantum technologies. *Submitted.* (2025) [arXiv:2509.10760](https://arxiv.org/abs/2509.10760)
- [22] H. Gao, P. Put, N. T. Leitao, N. U. Köylüoğlu, A. Maccabe, M. Mammen, S. Dandavate, **L. B. Hughes Wyatt**, L. S. Martin, A. C. Bleszynski Jayich, H. Park, M. D. Lukin, Nanoscale magnetometry via collective many-body dynamics in diamond. *Submitted* (2026)
- [23] L. Chen,* S. Parthasarathy,* S. A. Meynell,* **L. B. Hughes Wyatt**, E. Postelnicu, H. Yang, Z. Wang, W. Wu, W. V. Peloso, C. K. Kim, C. R. Laumann, K. Mukherjee, N. Y. Yao, A. C. Bleszynski Jayich. Engineering and Probing a One-Dimensional Dipolar Spin Ensemble in Diamond. *In preparation.*

- [24] T. Ono,* W. Wu,* H. Yang, L. B. Hughes Wyatt, B. Brenner, C. Liu, C. Fan, C. R. Laumann, J. N. Hallén, E. J. Davis, A. C. Bleszynski Jayich, N. Y. Yao. Direct Observation of Dipolar-Driven Anisotropic Quantum Projection Noise in a Solid-State Spin Ensemble. *In preparation*.
- [25] L. B. Hughes Wyatt, Y. Yan, A. A. Hossain, A. J. Minnich. Atomic layer etching of niobium via alternating O₂ and H₂/SF₆ plasma exposure. *In preparation*.

INVITED PRESENTATIONS

- **Materials Research Society, Spring Meeting**, Apr. 2026, Honolulu, HI, U.S.A.
- **American Physical Society Global Physics Summit**, Mar. 2026, Denver, CO, U.S.A.
- **Future Leaders Seminar Series**, Northwestern University Materials Department, Apr. 2024, Virtual.
- **Hasselt Diamond Workshop - SBDD XXVIII**, Feb. 2024, Hasselt, Belgium.
- **European Materials Research Society Meeting**, Jun. 2023, Strasbourg, France.

CONTRIBUTED PRESENTATIONS

- **Materials Research Society, Fall Meeting**, Dec. 2026, Boston, MA, U.S.A.
- **Materials Research Society, Spring Meeting**, Apr. 2025, Seattle, WA, U.S.A.
- **Materials Research Society, Fall Meeting**, Nov. 2023, Boston, MA, U.S.A.
- **Electronic Materials Conference**, Jun. 2023, Santa Barbara, CA, U.S.A.
- **New Diamond and Nano Carbon Conference**, Jun. 2023, East Lansing, MI, U.S.A.
- **Hasselt Diamond Workshop - SBDD XXVI**, Feb. 2022, Hasselt, Belgium.
- **Electronic Materials Conference**, Jun. 2020, Virtual.

TEACHING

- **Introduction to Quantum Mechanics (MATRL 200Q)** *Fall 2020*
Teaching Assistant for Professor Anton Van der Ven
- **Engineering Quantum Mechanics (MATRL 211B)** *Spring 2022 and Spring 2024*
Reader for Professor Chris Van de Walle

UNDERGRADUATE MENTORING

- **Isaac Kantor**, UCSB (2024-2025) *Awarded the Goldwater Scholarship (2026).*
- **Ekaterina Osipova**, UCSB (2021-2025)
- **Ian McGuire**, Quantum Foundry REU (Summer 2024) *Now Ph.D. student in Physics at Brown University.*
- **Melika Dabiri**, Quantum Foundry REU (Summer 2024)
- **Haopu Yang**, UCSB (2022-2023) *Now Ph.D. student in Physics at Harvard University.*

PROFESSIONAL SERVICE

- **Editorial and Peer Review**
Reviewer for Journal of Applied Physics, Journal of Materials Chemistry C, Advanced Quantum Technologies, Frontiers in Quantum Science and Technology, Applied Research.
- **Teaching Assistant - Cleanroom Training Program** *Aug. 2023 - Jan. 2024*
Central Coast Partnership for Regional Industry-focused Micro/Nanotechnology Education, UCSB 
 - Led cohorts of students in two 40 hour programs focused on wafer fabrication processes in the cleanroom.
 - Taught hands-on skills including photolithography, metal deposition, dry and wet etching, and device inspection/characterization.
 - Served primarily URM and community college students.
- **President, Treasurer, and Committee Member - Materials Scientist Association (MSA)** *Jan. 2021 - Sep. 2023*
Materials Department, UCSB 
 - Led student organization, planned prospective graduate student recruitment events, interfaced with department leadership.
 - Goals of promoting community and professional development for students in the Materials department.
 - Wrote a proposal and received grant funding for mental health-focused programming for students in the department.
- **Outreach Programs** *Jan. 2021 - Sep. 2024*
Quantum Foundry, Materials Research Lab, and Center for Science and Engineering Partnerships, UCSB 
 - Directed outreach project building Quantum Kits for middle school teachers.
 - Led a team of 10 students to develop materials and lesson plans for introducing students to quantum information science.
 - Premiered the developed activities at World Quantum Day and in local middle schools as part of the Family Science Ultimate Exploration program.
 - Participated in other volunteering and outreach programs at local elementary schools.
- **Writing Consultant** *Sep. 2021 - Nov. 2022*
Graduate Writing Center, UCSB 
 - Reviewed application essays through individual consultations with students applying for the NSF GRFP and NDSEG fellowships.
 - Provided tailored feedback as well as general strategies for successful proposal writing. Supervisor: Dr. Robby Nadler.

REFERENCES

1. **Ania C. Bleszynski Jayich**
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University of California Santa Barbara
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Relationship: Ph.D. Thesis Advisor
2. **Kunal Mukherjee**
Assistant Professor, Department of Materials Science and Engineering
Stanford University
Email: kunalm@stanford.edu
Relationship: Ph.D. Thesis Co-Advisor
3. **Austin J. Minnich**
Professor, Department of Mechanical Engineering and Applied Physics
California Institute of Technology
Email: aminnich@caltech.edu
Phone: +1-626-395-2142
Relationship: Postdoc Advisor
4. **Norman Y. Yao**
Professor, Department of Physics
Harvard University
Email: nyao@fas.harvard.edu
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Relationship: Collaborator